Last time

- Mid term recap
- Newton - Euler dynamics
- Quadrotor example

Today

- Warm-up (with another recap :)
- Airplane example (Newton-Euler)
- Airplane simulator

RECAP

Newton's laws

3 laws ⟶ 2nd law $\boxed{\frac{dp}{dt} = F}$ $p = mv$

Fundamental balance of linear momentum

Momentum conservation

$$F = 0 \rightarrow \frac{dp}{dt} = 0 \Rightarrow p \equiv cte \equiv \text{Momentum conser}$$

Energy

$$\frac{dp}{dt} = F_{ext}$$

$$m\dot{v} = F$$

$$m\dot{v}v^T = Fv^T$$

$$\frac{d}{dt}\underbrace{\left(\frac{1}{2}m\|v\|^2\right)}_{T} = Fv^T$$

(kinetic energy)

$$\dot{T} = P$$

(power)

$$P \rightarrow \underbrace{Fv^T}$$

$$W = \Delta T \rightarrow \text{Energy change that is transferred through F}$$

$$P = \frac{dW}{dt} \rightarrow \text{The change in energy rate}$$

$$\dot{T} = Fv^T \rightarrow \text{conservative force} \Rightarrow \begin{cases} U(r) \equiv \text{ener potenti} \\ F = -\nabla U \end{cases}$$

$$\dot{T} = -\nabla U v^T$$

$$\dot{T} = -\dot{U}$$

$$U(x(t), y(t), z(t))$$

$$\frac{dU}{dt} = \nabla U v^T$$

$$\frac{d}{dt}(T+U) = 0$$

$$T + U \equiv cte \equiv \text{Energy conservation}$$

Eqs. of motion

- Two strategies
  1. Euler - Lagrange
  2. Newton - Euler

① Euler - Lagrange ⇒ energy and generalized coordinates

$$L = T(q, \dot{q}) - U(q)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = F_i$$

② Newton - Euler ⇒ forces and moments

$$x = \begin{bmatrix} {}^N r & {}^N q^B & {}^B v & {}^B w \end{bmatrix}$$

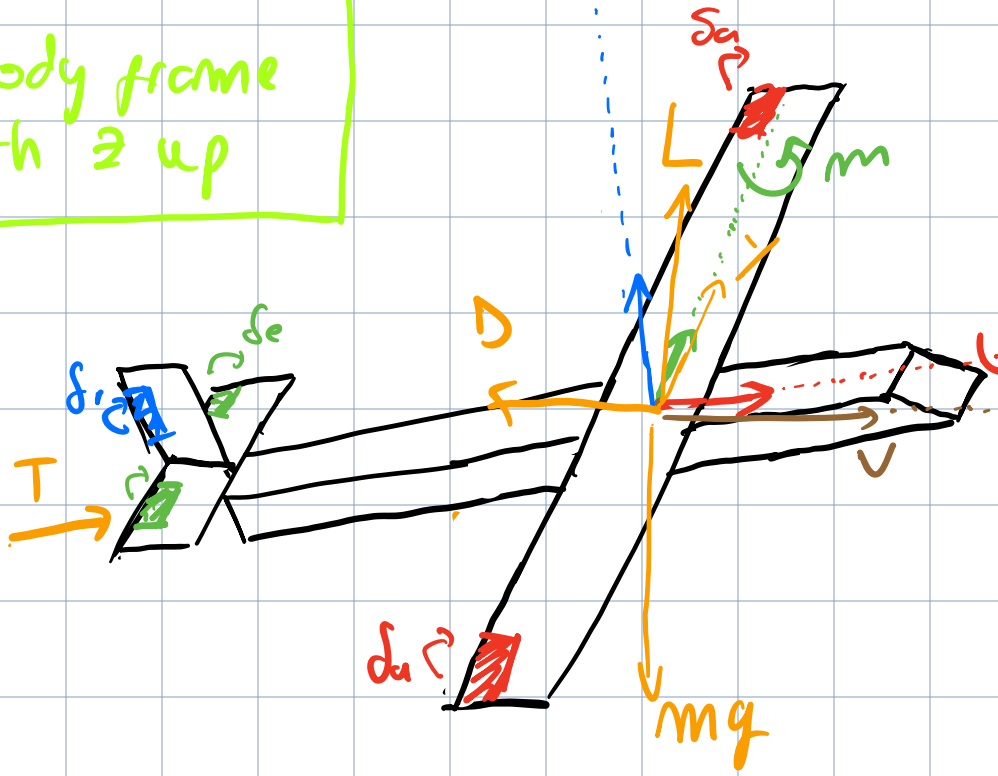
$$\dot{x} = \begin{bmatrix} Q^B v \\ \frac{1}{2} G(q)\, {}^B w \\ \frac{1}{m}\, {}^B F - {}^B w \times {}^B v \\ J^{-1}\left({}^B \tau - {}^B w \times J\, {}^B w \right) \end{bmatrix}$$

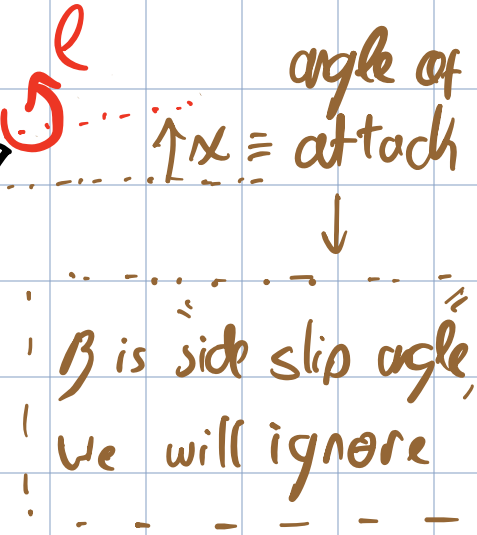
- These eqs. of motion are in body frame !!!
- Can model lots of systems

EXAMPLE: AIRPLANE

- We can reuse Newton-Euler
- We "only" need to find ${}^{B}F$ and ${}^{B}Z$

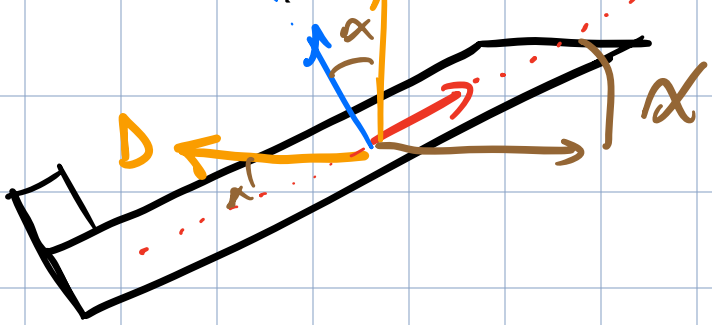
⚠ Body frame with $\hat{z}$ up





Longitudinal forces

$$ {}^{B}F = \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix} + Q^{T} \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + \begin{bmatrix} L\sin\alpha - D\cos\alpha \\ 0 \\ L\cos\alpha + D\sin\alpha \end{bmatrix} $$



Angular forces (torques)

$$^{B}Z = \begin{bmatrix} l \\ m \\ n \end{bmatrix}$$

- l → x axis → roll → aileron (δa)
- m → y axis → pitch → elevator (δe)
- n → z axis → yaw → rudder (δr)

Aerodynamic forces

$$\bar{q} = \frac{1}{2} \rho \|V\|^2$$

- ρ → air density
- $\bar{q}$ → dynamic pressure

S → wing area

$$\begin{cases} L = \bar{q} S C_L \\ D = \bar{q} S C_D \\ Y = \bar{q} S C_Y \end{cases}$$

b → wing span, c → chord length

$$\begin{bmatrix} l \\ m \\ n \end{bmatrix} = \bar{q} S \begin{bmatrix} b\, C_l \\ c\, C_m \\ b\, C_n \end{bmatrix}$$

Aerodynamic coefficients

Coefficient	Effect	Dependencies
C_L	Lift	α, w, δ_e
C_D	Drag	α, δ_e
C_Y	Side force	$\alpha, \beta, \delta_a, \delta_r$
C_ℓ	Roll	$\beta, w, \delta_a, \delta_r$
C_m	Pitch	α, w, δ_e
C_n	Yaw	$\beta, w, \delta_a, \delta_r$

- These dependecies are only valid for conventional low-speed flight
- Appropriately modeling these requires high fidelity models